$$\vec{F} = [x, y, z]$$

(a)

$$\begin{array}{lcll} x = r\cos\theta & u = r & x = u\cos v & 0 \le u \le 2 \\ y = r\sin\theta & \longrightarrow & y = u\sin v & 0 \le v \le 2\pi \\ z = z & v = \theta & z = 4 - u^2 & \end{array}$$

$$\vec{r}(u,v) = [u\cos v,\ u\sin v,\ 4 - u^2]$$

$$\vec{r}_u = [\cos v,\ \sin v,\ -2u]$$

$$\vec{r}_v = [-u\sin v,\ u\cos v,\ 0]$$

$$\vec{N} = \vec{r}_u \times \vec{r}_v = \left[2u^2\cos v,\ 2u^2\sin v,\ \overbrace{u\cos^2 v + u\sin^2 v}^{u}\right]$$

$$F(\vec{r}(u,v)) = [u\cos v,\ u\sin v,\ 4 - u^2]$$

$$\begin{aligned} \vec{F} \cdot \vec{N} &= 2u^3\cos^2 v + 2u^3\sin^2 v + 4u - u^3 \\ &= 2u^3 + 4u - u^3 \\ &= u^3 + 4u \end{aligned}$$

$$\begin{aligned} \iint_S \vec{F} \cdot d\vec{A} &= \iint_R \vec{F} \cdot \vec{N}\, du\, dv \\ &= \int_0^{2\pi}\int_0 u^3 + 4u\, du\, dv \\ &= 2\pi \left(\frac{u^4}{4} + 2u^2 \Big|_0 \right) \\ &= 2\pi\,(4 + 2 \cdot 2^2) = 24\pi \end{aligned}$$

(b)

$$\begin{array}{lcl} u = x & & x = u \\ v = y & \longrightarrow & y = v \\ & & z = 4 - 4u - 8v \end{array}$$

$$\vec{r}(u,v) = [u, v, 4 - 4u - 8v]$$

$$\vec{r}_u = [1, 0, -4]$$

$$\vec{r}_v = [0, 1, -8]$$

$$\vec{N} = \vec{r}_u \times \vec{r}_v = [4, 8, 1]$$

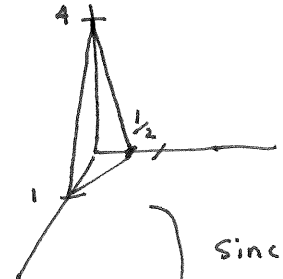
$$F(\vec{r}(u,v)) = [u, v, 4 - 4u - 8v]$$

$$\vec{F} \cdot \vec{N} = \cancel{4u} + \cancel{8v} + 4 - \cancel{4u} - \cancel{8v} = 4$$

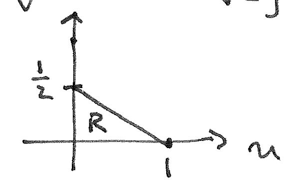
$$\iint_S \vec{F} \cdot d\vec{A} = \iint_R \vec{F} \cdot \vec{N} \, du\,dv$$

$$= \iint_R 4 \, du\,dv$$

$$= 4\left(\tfrac{1}{2}(1)\left(\tfrac{1}{2}\right)\right) = 1$$



Since $u = x$ & $v = y$



Area(R) $= \frac{1}{4}$

Eqn of Line $v = -\frac{1}{2}u + \frac{1}{2}$

(c)

$$\begin{aligned} x &= \rho \sin\phi \cos\theta \\ y &= \rho \sin\phi \sin\theta \\ z &= \rho \cos\phi \end{aligned} \quad \xrightarrow[v=\theta]{\rho = 2,\ u = \phi} \quad \begin{aligned} x &= 2\sin u \cos v \\ y &= 2\sin u \sin v \\ z &= 2\cos u \end{aligned}$$

$$\vec{r}(u,v) = [2\sin u \cos v,\ 2\sin u \sin v,\ 2\cos u]$$

$$\vec{r}_u = [2\cos u \cos v,\ 2\cos u \sin v,\ -2\sin u]$$

$$\vec{r}_v = [-2\sin u \sin v,\ 2\sin u \cos v,\ 0]$$

$$\vec{N} = \vec{r}_u \times \vec{r}_v = \left[4\sin^2 u \cos v,\ 4\sin^2 u \sin v,\ \overbrace{4\sin u \cos u \cos^2 v + 4\sin u \cos u \sin^2 v}^{4\sin u \cos u}\right]$$

$$\vec{F}(\vec{r}(u,v)) = [2\sin u \cos v,\ 2\sin u \sin v,\ 2\cos u]$$

$$\begin{aligned} \vec{F} \cdot \vec{N} &= 8\sin^3 u \cos^2 v + 8\sin^3 u \sin^2 v + 8\sin u \cos^2 u \\ &= 8\sin^3 u + 8\sin u \cos^2 u \\ &= 8\sin u(\sin^2 u + \cos^2 u) = 8\sin u \end{aligned}$$

$$\iint_S \vec{F} \cdot d\vec{A} = \iint_R \vec{F} \cdot \vec{N} \, du\,dv = \int_0^{2\pi}\int_0^{\pi} 8\sin u \, du\,dv$$

$$= 2\pi\left(-8\cos u \Big|_0^{\pi}\right)$$

$$= -16\pi(-1-0) = 32\pi$$